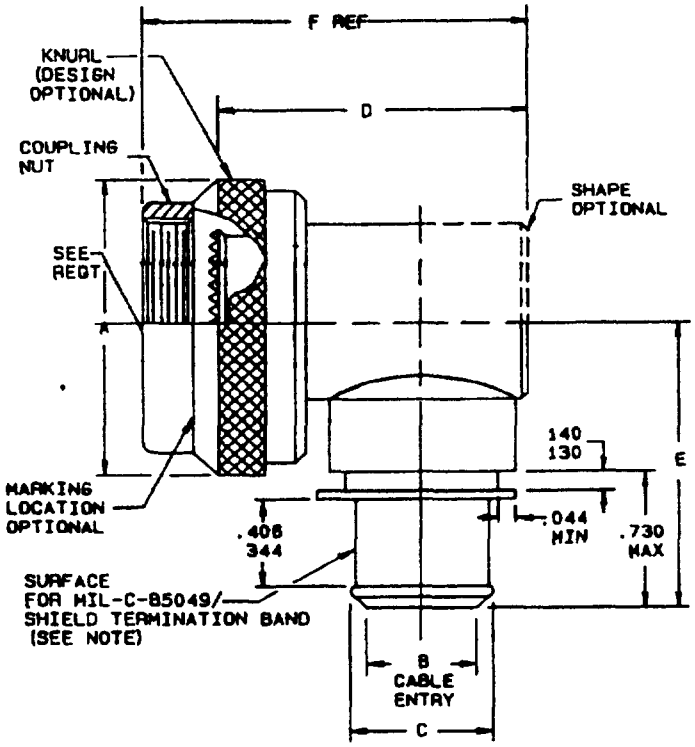
MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, 90°, SELF-LOCKING, SHIELD BAND TERMINATION, (RFI/EMI), SHRINK SLEEVE ACCOMMODATION, CATEGORY 38 (FOR MIL-C-38999 SERIES III AND IV CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-85049



NOTE Ribbed, internal knurl or surface roughness R_a 125-250 in accordance with ANSI B46 1 See figure 2 for selective plating details

FIGURE 1. Configuration and dimensions

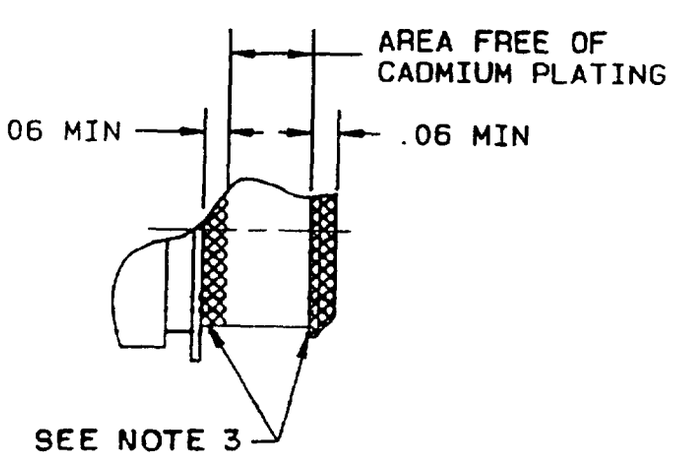
DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
044	1 12	.500	12 70	770	19 56	1 083	27 51	1 516	38.51	1 773	45 03
130	3 30	562	14 27	812	20 62	1 145	29 08	1 553	39 45	1 793	45 54
140	3 56	583	14 81	829	21 06	1 157	29 39	1 562	39 67	1 859	47 22
250	6 35	625	15 88	858	21 79	1 279	32 49	1 614	41.00	1 875	47 62
312	7 92	660	16 76	895	22 73	1 375	34 92	1 642	41 71	1 889	47 98
395	10 03	684	17 37	938	23 83	1 404	35 66	1 678	42 62	1 919	48 74
438	11 13	707	17 96	957	24 31	1 406	35 71	1 687	42 85	1 938	49 22
457	11 61	750	19 05	984	24 99	1 437	36 50	1 750	44 45	2 062	52 37
				1 000	25 40	1 467	37 26	1.768	44 91	2 125	53 98

NOTES.
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 00 (25 4 mm)
3. Cadmium/nickel interface shall be coated with polysulfide sealant

FIGURE 2 Selective plating



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TABLE I Shell size and dimensions

Accessory shell size	Shell size code (ref)	A max dia	B dia ±.010 entry size		C dia ref		D max dim	E max dim	F ref
			02	03	02	03			
9	A	.858	N/A	250	N/A	395	1 375	1 417	1 623
11	B	.984	N/A	.312	N/A	.457	1.437	1.480	1 685
13	C	1.157	.312	.438	457	.583	1.562	1.553	1 81
15	D	1.279	438	.562	583	.707	1.687	1.614	1.935
17	E	1 406	500	625	.645	770	1.750	1 678	1.998
19	F	1.516	.625	.750	770	895	1 875	1 773	2 123
21	G	1.642	625	812	770	957	1 938	1 796	2 186
23	H	1 768	.688	938	829	1.083	2.062	1 859	2.31
25	J	1 889	750	1 000	895	1.145	2.125	1 919	2.373

REQUIREMENTS.

Design and construction

Dimensions and configuration See figure 1 and table I

Interface dimensions: Shall conform to MIL-C-85049

Coupling nut Shall be captivated to the adapter body and free to rotate

Material and finish In accordance with MIL-C-85049 and table II

TABLE II Material and finish

Finish	Material
G - Electroless nickel 1/ (space grade) N - Electroless nickel 1/ P - Cadmium olive drab over electroless nickel (selective plating, see figure 2) W - Cadmium olive drab	Aluminum alloy
J - Cadmium olive drab L - Cadmium olive drab over electroless nickel (selective plating, see figure 2) 2/ M - Electroless nickel	Composite

1/ Not for Navy use Limited to space applications for Air Force use
2/ Must meet the requirements, and pass the tests for finish J.

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Vibration Backshells shall be subjected to test parameters of applicable connector specifications The coupling torque shall be within +20, -10 inch-pounds of the initial value

Additional tests for finishes J, L, and M shall be as follows:

Ozone exposure. Backshells shall be tested in accordance with MIL-STD-1344, method 1007

Sample Each sample shall consist of one backshell

Sample preparation: Not applicable

Failure: Blistering, or peeling of the plating constitutes a failure Any condition which adversely affects the function of the backshell also constitutes a failure

Fluid immersion (see table III): Backshell shall be tested in accordance with MIL-STD-1344, method 1016

Sample: Each sample shall consist of one backshell

After immersion backshells shall meet the coupling thread strength requirements of MIL-C-85049.

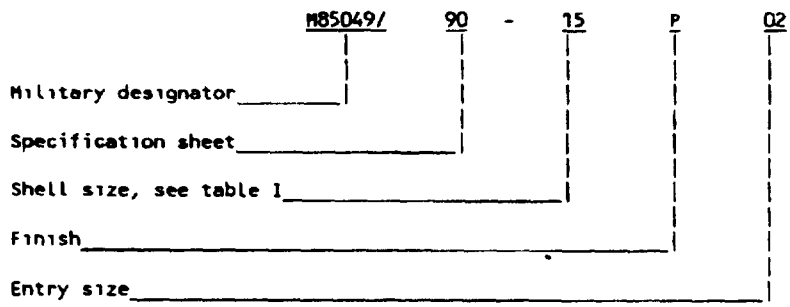
TABLE III Test fluids

Sample number	Test fluid (method 1016 of MIL-STD-1344)
1	d
2	e
3	a
4	b
5	f
6	g
7	c
8	l
9	h
10	i
11	j
12	k

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Thermal vacuum outgassing (finish G) Backshell shall not emit vacuum condensable, noxious or toxic gasses when tested in accordance with ASTM E595 Total mass loss (TML) shall not be greater than 1 0 percent and collected volatile condensable materials (CVCM) shall not be greater than 10 percent Data listed in NASA Reference Publication 1124 Revised may be used in lieu of actual test data for application materials.

Part or Identifying Number (PIN) See example below



QUALIFICATION

Qualification See MIL-C-85049, category 3B

QPL evaluating activity: Naval Air Warfare Center, Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities
Army - AR, MI
Navy - EC, MC, OS
Air Force - 11, 15, 19, 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5935-3760-09)

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2750 LOG/ES
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TELEPHONE - COM 513-296-5517 DSN 986-5517
FAX - COM 513-296-8783 DSN 986-8783

VERY DISSATISFIED	DISSATISFIED	SOMEWHAT DISSATISFIED	SOMEWHAT SATISFIED	SATISFIED	VERY SATISFIED	NOT APPLICABLE
1	2	3	4	5	6	7

(PLEASE ENTER THE VALUE THAT YOU FEEL IS APPROPRIATE)

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III. COMMENTS/SUGGESTIONS

(CONSTRUCTIVE COMMENTS OR SUGGESTIONS APPRECIATED)

MIL-C-85049/90

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